

Degrees of String Admissibility

The Chomsky hierarchy, context-freeness, regularity, and linearity

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A language family is a degree of admissible dependency

Question

How much structure must a description preserve to distinguish the intended strings?

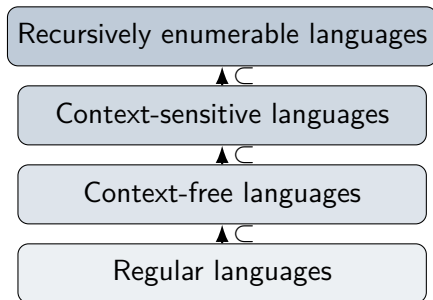
- Formal languages are sets of strings.
- Families classify the descriptions that can characterize those sets.
- Greater descriptive capacity admits more dependencies—and more unwanted analyses.

MOLR §§3.1–3.3

Where we are

1. The hierarchy
2. Admissibility tests
3. Interpretation

The classical hierarchy is a chain of set inclusion



Take-away

The hierarchy classifies stringsets by possible descriptions, not languages by cultural or cognitive “complexity.”

Natural-language work cuts the hierarchy at finer seams

regular $\subset$ linear $\subset$ context-free $\subset$ linear-indexed $\subset$ context-sensitive

- Linear languages refine the space between regular and context-free descriptions.
- Linear-indexed languages refine the space above context-free descriptions.
- These additions target kinds of dependency, not prestige or overall language complexity.

Guided analysis

Why is “English is context-free” a poorly formed empirical claim?

Where we are

1. The hierarchy
2. Admissibility tests
3. Interpretation

A pumping lemma states a necessary property of a family

For sufficiently long $z \in L$, a decomposition must permit controlled repetition.

What the argument can show

If every permitted decomposition fails, then L is not in that family.

What it cannot show

Finding one pumpable string does not prove that L belongs to the family.

Regular admissibility permits one pumped region

If L is regular, sufficiently long $z \in L$ can be written

$$z = uvw$$

such that

$$|v| \geq 1, \quad |uv| \leq n, \quad uv^i w \in L \text{ for every } i \geq 0.$$

Guided analysis

Why does $\{a^n b^n \mid n \geq 0\}$ resist a single locally pumped region?

MOLR §3.2, regular admissibility

Context-free admissibility permits two coordinated regions

If L is context-free, sufficiently long $z \in L$ can be written

$$z = uvwxy$$

such that

$$|vx| \geq 1, \quad |vwx| \leq n, \quad uv^iwx^iy \in L \text{ for every } i \geq 0.$$

- Two substrings can grow together.
- Their contribution is structurally linked through a derivation.

Linear admissibility restricts where the paired growth occurs

For a linear language, write sufficiently long z as $uvwxy$, with

$$|vx| \geq 1, \quad |uvxy| \leq n, \quad uv^jwx^jy \in L.$$

- The paired material lies toward the two edges of the derivational dependency.
- Only one nonterminal continues the dependency at each step.

Question

What structural contrast might this capture in morphology?

MOLR §3.2, linear admissibility

Where we are

1. The hierarchy
2. Admissibility tests
3. Interpretation

Admissibility is structural without reifying a machine

strings $\longrightarrow$ invariant decomposition property $\longrightarrow$ degree of structure

- A pumping property characterizes what must remain possible across variants.
- It does not assert that speakers literally pump substrings.
- It prepares the ground for grammars that describe and automata that compute.

Take-away

First classify the dependency; later choose a method that realizes it.

MOLR §3.3

Third hour: choose the weakest adequate family

Consider these languages:

$$L_1 = \{a^n b^m \mid n, m \geq 0\}, \quad L_2 = \{a^n b^n \mid n \geq 0\}, \quad L_3 = \{ww \mid w \in \{a, b\}^*\}.$$

- ① What dependencies must each description preserve?
- ② Which family is clearly too weak?
- ③ What evidence would be needed for a positive classification?

Week 5 synthesis

- 1 Families classify formal languages by descriptive capacity.
- 2 Inclusion gives a scale, not a claim about whole natural languages.
- 3 Pumping lemmas provide necessary conditions and negative tests.
- 4 Regular, linear, and context-free admissibility preserve different dependency patterns.
- 5 Classification precedes the choice of grammar or machine.

Question

Quiz 4 asks you to diagnose a small string pattern and justify the diagnosis.